



Bazar-Trace: Offline-First PWA Stock & Inventory Management System

Built with HTML, CSS, Vanilla JS, Node.js (Custom HTTP Server) & Oracle Database (Local)

> Presenter_Metadata

Name: Arif Hossen

ID: 231010687

Dept: Computer Science & Engineering (CSE)

> Supervision_Metadata

Supervisor: Reduanul Bari Subhon sir

Mentor: Israt Jahan miss

What is Bazar-Trace?



Vanilla SPA

Built purely with raw ES Modules, HTML5, and CSS variables—zero heavy frameworks.



Offline-First Sync Engine

Records transactions and looks up local product data even during complete network drops.



Role-Based Security

Strict functional separation between admin management interfaces and staff operational views.



Mobile-First Design

Hyper-optimized interface for low-end retail devices and simple mobile browsers.

Inventory Dashboard

Stock Level



.... Pending Sync

Recent Sales

10:45 AM - Order #12345	\$120.50
10:30 AM - Order #12344	\$55.00
10:15 AM - Order #12343	\$89.99

Scan Barcode

MOTIVATION

The Catalyst: Why Cloud ERPs Fail Local Retail

	Legacy Cloud ERPs	Bazar-Trace Local PWA
Network Reliance	✗ Fails during Bangladesh's frequent internet instability.	✓ 100% operational offline via Service Workers.
Operational Cost	✗ Requires expensive desktop hardware and heavy recurring software licenses.	✓ Runs on existing low-end mobile devices; zero licensing fees.
UX Complexity	✗ Cluttered, complex portals built for accountants.	✓ Minimalist, cashier-focused interface with automated barcode scanning.
Academic Constraints	✗ Relies on massive third-party libraries and cloud databases.	✓ Built entirely from scratch using raw JS and a local Oracle DB constraint.

Core Project Objectives



Zero Latency Downtime

Enable cashiers to record Sales (OUT) and Stock received (IN) instantly, regardless of network state.



Conflict-Free Auto Synchronization

Replay offline mutations idempotently and in strict FIFO (First-In-First-Out) order upon network recovery.



Native Hardware Integration

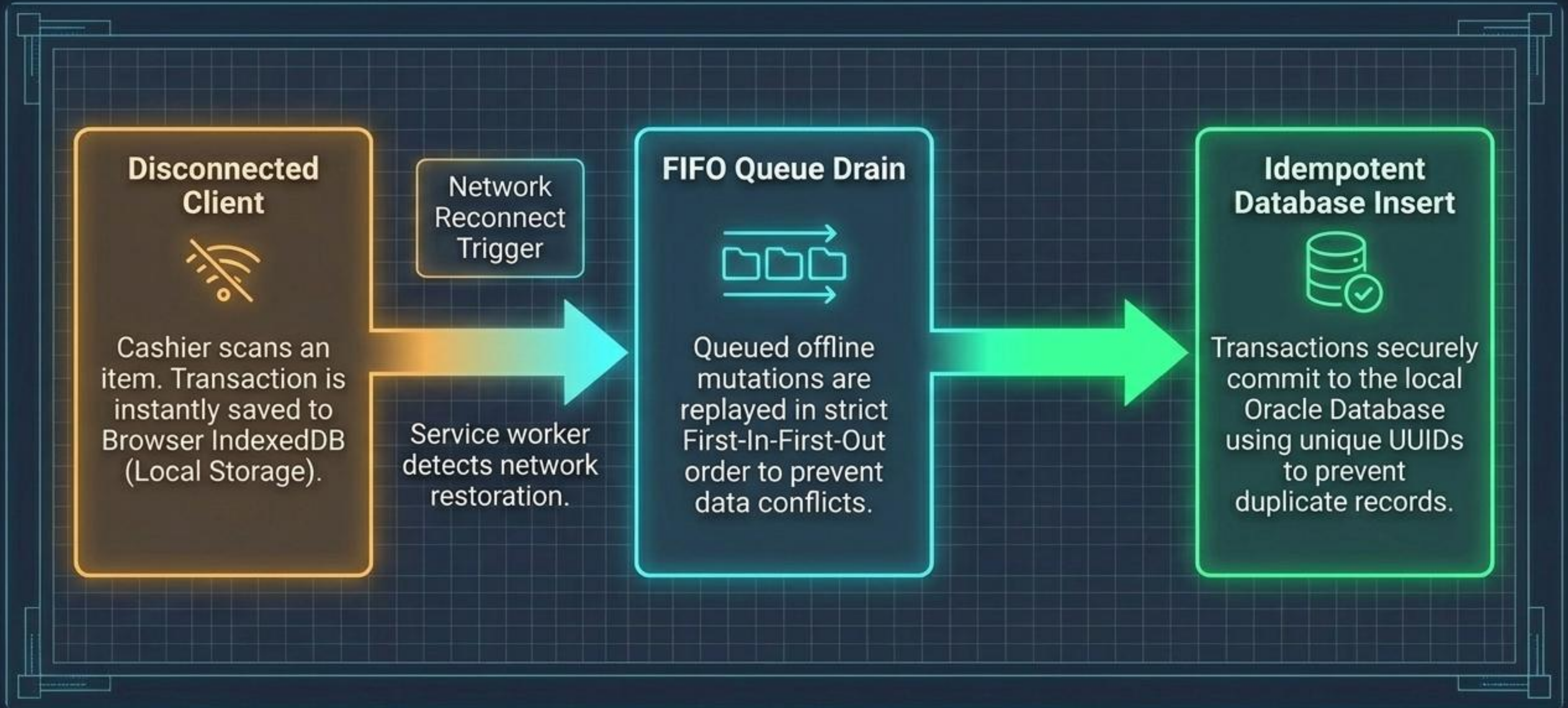
Access native camera modules for barcode scanning and trigger synthesized Web Audio scan beeps without external apps.



Resource Constraints Compliance

Deliver a production-grade system strictly adhering to university guidelines: Vanilla CSS, raw JavaScript, and Oracle DB.

The Offline-First Sync Engine



The “Zero-Framework” Tech Stack

Frontend Core

HTML5, CSS Variables, ES6 JavaScript Modules.

- Web MediaDevices (getUserMedia) & Web Audio API (Synthesizer).

PWA Engine & Backend

Service Worker (caching) + Browser IndexedDB.

- Node.js (ESM) Custom HTTP server: hand-built router, middleware, body parser, CORS handler (Zero Express).

Database & Infra

Oracle Database XE 21c (Local Machine).

- oracledb v6 Thin mode (No Instant Client).
Docker Compose for production.



Role-Based Functional Boundaries

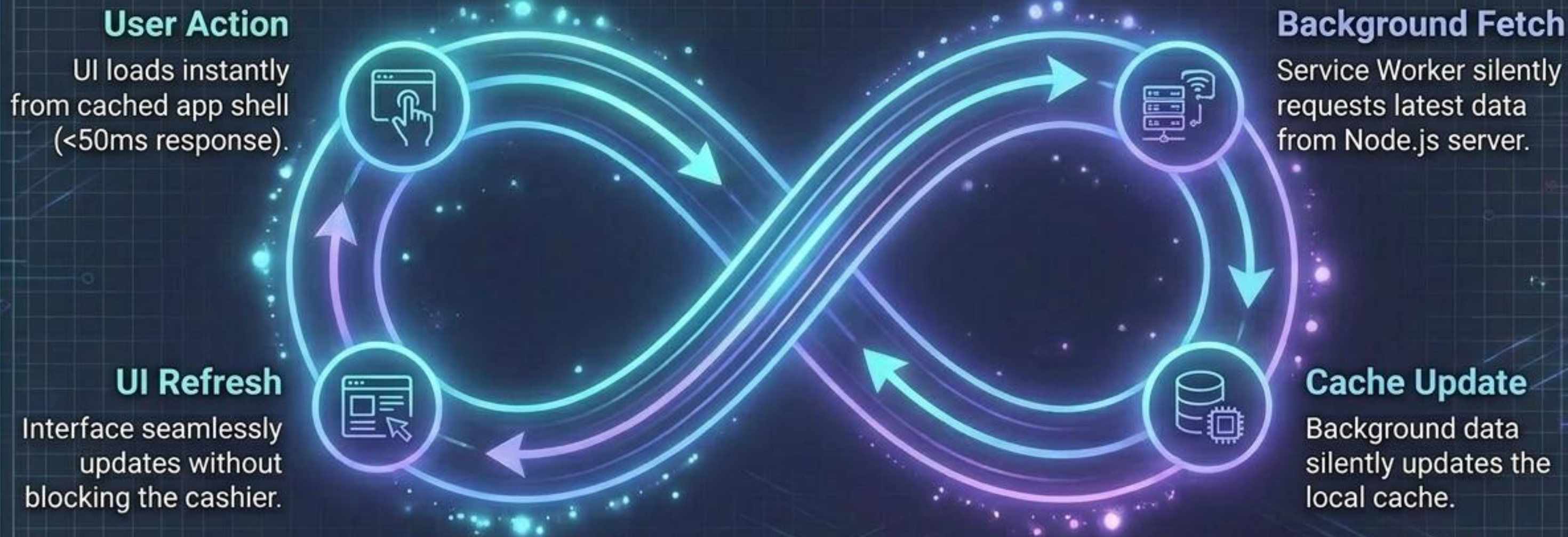
Cashier & Staff (Operational)

- ✓ Secure Login & Session validation
- ✓ Log transactions (Sale OUT / Stock Received IN)
- ✓ Dynamic autocomplete product search
- ✓ Native camera barcode autofill

Administrator (Management)

- ✓ Full Product CRUD (Create, Edit, Delete)
- ✓ Set custom stock depletion alerts
- ✓ Staff Account Registry & active status toggling
- ✓ Dashboard Analytics: Live stock counts, canvas-drawn sales graphs, and expiry warnings

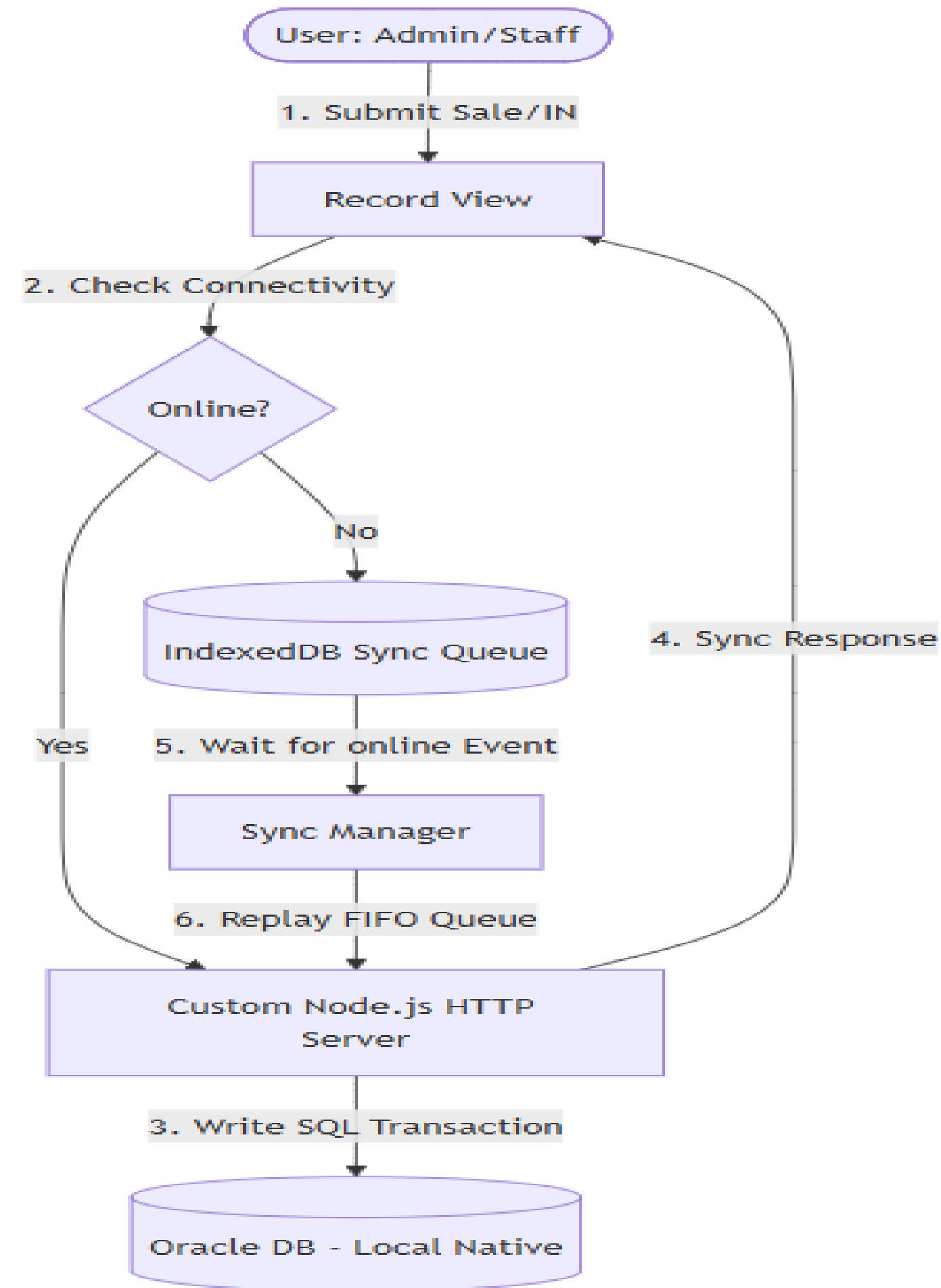
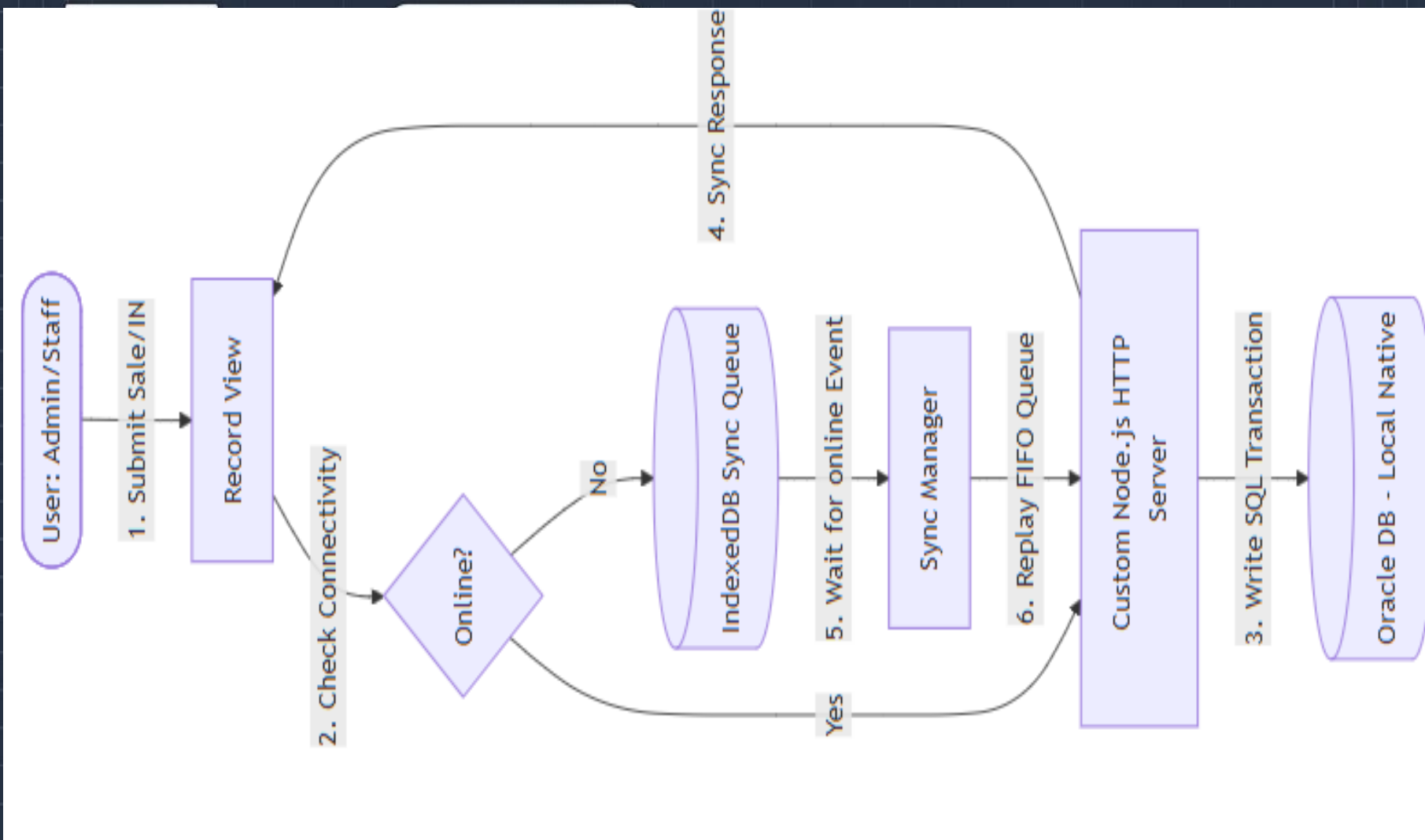
System Performance, Reliability, and Security Baselines



Reliability: Client UUID checks prevent duplicate inserts.
Security: Cryptographic Bcrypt hashing + JWT authentication.

Data Flow Diagram (Level-1)

Transaction Data Flow Across the Network Boundary

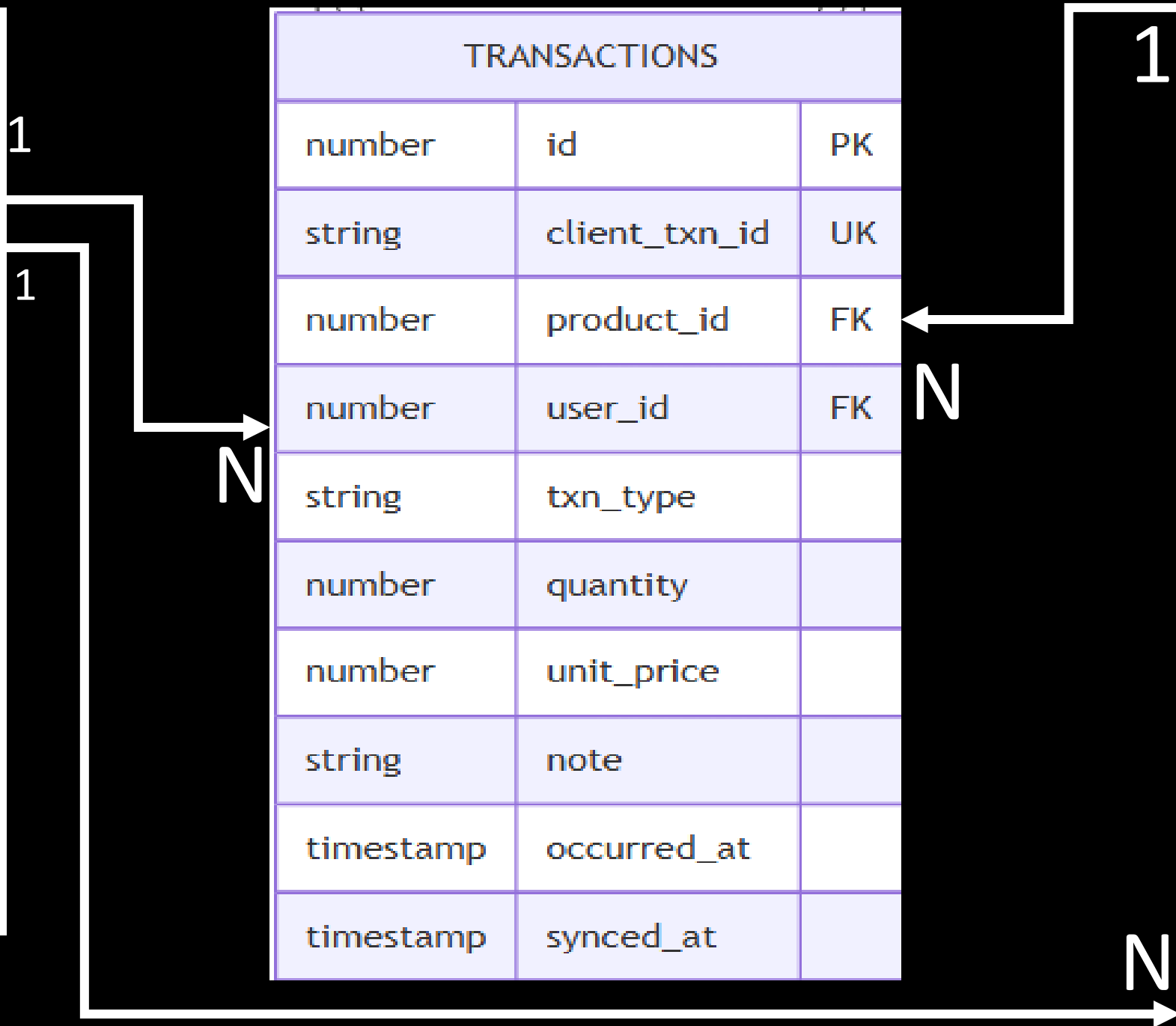


Relational Database Architecture

USERS		
number	id	PK
string	email	UK
string	password_hash	
string	full_name	
string	role	
boolean	is_active	
timestamp	created_at	

TRANSACTIONS		
number	id	PK
string	client_txn_id	UK
number	product_id	FK
number	user_id	FK
string	txn_type	
number	quantity	
number	unit_price	
string	note	
timestamp	occurred_at	
timestamp	synced_at	

PRODUCTS		
number	id	PK
string	sku	UK
string	barcode	
string	name	
string	category	
string	unit	
number	stock	
number	cost_price	
number	sell_price	
number	low_stock_threshold	
date	expiry_date	
number	created_by	FK



Executing the Schema in Oracle Database

```
CREATE TABLE transactions (  
    tx_uuid VARCHAR2(36) PRIMARY KEY,  
    user_id NUMBER NOT NULL,  
    product_id NUMBER NOT NULL,  
    tx_type VARCHAR2(10) CHECK (tx_type IN ('IN', 'OUT')),  
    quantity NUMBER NOT NULL,  
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    sync_status VARCHAR2(15) DEFAULT 'PENDING',  
    CONSTRAINT fk_user FOREIGN KEY (user_id) REFERENCES users(user_id),  
    CONSTRAINT fk_product FOREIGN KEY (product_id) REFERENCES products(product_id)  
);
```

Structured Software Development Life Cycle



System Resilience (SWOT Analysis)

Strengths

- 100% offline uptime via IndexedDB/Service Workers;
- Ultralight load sizes (zero JS framework overhead);
- Conflict-free transactions via client UUIDs.

Weaknesses

- Database updates fundamentally rely on client synchronization triggers;
- Camera barcode detection depends on modern browser support.

Opportunities

- Expansion into multi-terminal retail point-of-sale setups;
- Integration with local hardware receipt printers.

Threats

- Device clear-cache commands could inadvertently wipe unsynced offline data;
- Evolving browser security standards for native camera feeds.

The Business Case: Cost vs. Value Delivered

Design & UI/UX:	10k-15k BDT
Frontend Dev:	45k-60k BDT
Backend Dev:	60k-80k BDT
Testing & QA:	15k-25k BDT
Domain/Hosting: (Bypassed via Local Dev)	5k-10k BDT
Oracle Database: (Bypassed via Oracle XE 21c Free Local)	12k-24k BDT

Market Estimate:
147,000 - 214,000 BDT.

**Actual
Solo Cost:**
0 BDT.

Validating the Power of Offline-First Web Technologies

- ✓ Bazar-Trace operates as a fully functional, offline-resilient inventory application tailored for low-connectivity retail.
- ✓ Strictly satisfies all academic requirements by exclusively utilizing pure vanilla code and relational Oracle schemas.
- ✓ Successfully demonstrates how modern native Web APIs can seamlessly replace heavy, expensive commercial frameworks.

Thank You / Q&A

Bazar-Trace Defense

University of Scholars | Student ID: 231010687

Supervisor: Reduanul Bari Subhon